

FIELDNODE · REQUIREMENTS

FieldNode requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/fieldnode		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-09-30	Amish Chadha	Design for construction (FND-DDR-003) applied; mass, cost, bracket, mounting and service figures updated; open for Amish's review
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from FND-CAL-001; R11 and R16 restated to match FND-DDR-001 (D5 and the FieldNode core costing)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status against each
0.1	2026-09-25	Amish Chadha	Initial scaffold

FieldNode requirements

Fifteen of the eighteen requirements are met on paper or by design, two are at risk, none is missed and one can only be verified by a timed installation (FND-CAL-001 v0.3, Table 4). Amish accepted the TRL 3 recommendations on 2026-09-25 (FND-DDR-002), and three targets are restated to match: R3 now requires the sun shield where the site's design maximum ambient exceeds 30 °C, R9 applies on The Things Network with a firmware rule that lengthens the interval at slow spreading factors, and R14 applies to the base node without the hot-climate shield. R7 is unchanged at 100 mW, which is now the published allowance. R11 and R16 stay as restated in v0.3 to match FND-DDR-001. The status column gives the standing from FND-CAL-001; "Met on paper" means shown by calculation, not by test.

Table 1. Requirements and status at TRL 3

ID	Requirement	Target	Verification (TRL 3 or later)	Status (FND-CAL-001)
R1	Weather protection	Enclosure IP65 or better with membrane vent; all penetrations sealed glands or rated connectors	Enclosure rating; later spray test	Met by design: IP65 box, ePTFE vent, two IP68 glands, capped IP67 ports and bulkhead
R2	Operating temperature	-20 to +45 °C ambient; electronics rated -20 to +70 °C inside the enclosure	Datasheets; thermal calculation	At risk (cold): warm end covered (52.2 °C dusty worst case with the shield at 45 °C; 58.3 °C unshielded at 30 °C); no charging on clear days colder than about -13 °C
R3	Interior temperature in sun	60 °C or less inside the enclosure in full sun, still air, at the site's design maximum ambient up to 45 °C; the ventilated sun shield (BOM line 14) is fitted wherever that maximum exceeds 30 °C	Thermal calculation; later test in sun	Met on paper: with the shield at 45 °C, 48.5 °C clean, 50.4 °C dusty, 52.2 °C worst sun position; unshielded at 30 °C, 58.3 °C worst sun position; shield factor assumed
R4	Safe charging window	Cell charging blocked below 0 °C and above 45 °C by an NTC on the cell	Charger design review	Met by design
R5	Energy neutral in the worst month	Daily harvest exceeds full-load demand at 2 peak sun hours	Energy budget	Met on paper: 7.75 Wh stored against 2.67 Wh drawn in the worst month; 10.8 to 13.1 Wh stored on a hot clear day with the shield; 9 V class panel keeps 2.42 V above the charger minimum input when hot
R6	Autonomy without sun	5 days or more at full sensor allowance, starting from 80 % charge usable	Energy budget	At risk (cold or aged cell): 5.75 days at the published 100 mW, a 15 % margin; 4.03 days at -20 °C and 4.60 days at end of life
R7	Sensor power allowance	100 mW average or more available to the sensor ports; 100 mW is the published allowance	Energy budget	Met on paper: 100 mW published; 115.0 mW is the ceiling for exactly 5 days
R8	Radio link	LoRaWAN uplink to a gateway 2 km away in suburban terrain at SF9 or faster	Link budget; later field test	Met on paper: 18.6 dB margin at 2 km to a 30 m gateway
R9	Airtime within fair use	30 s uplink airtime per day or less on The Things Network; firmware keeps the 15 min default at SF7 to SF9 and lengthens the interval at SF10 and slower (22 min at SF10, 48 min at SF11, 87 min at SF12)	Airtime calculation; later firmware review	Met on paper: 23.7 s at SF9; 30.0 s or less at every spreading factor under the rule
R10	Store and forward	30 days or more of readings kept on the node when the link is down	Memory calculation	Met on paper: 30 days in 90 kB of 16 MB
R11	Standard sensor interface	Two sealed M12 5-pin ports, each carrying I2C, UART or RS-485, one analog input and one switched rail		Met on paper; pinout open (FND-DDR-001, O2)

ID	Requirement	Target	Verification (TRL 3 or later)	Status (FND-CAL-001)
		selectable at 3.3, 5 or 12 V; a gland for fixed-cable sensors	Pinout review with adopting projects	
R12	Mounting and install	Fits 40 to 60 mm poles and flat walls; one person installs in 15 min or less with hand tools	Design review; later timed install	Fit met by design (seats 40 to 71 mm); time estimated at 15 min, not verifiable at TRL 3
R13	Wind	Survives 35 m/s (126 km/h) gusts on the panel without loosening	Load calculation	Met on paper: clamp pull 59 N against 2,000 N; band preload assumed
R14	Mass	2.5 kg or less for the base node, including panel and mounts; the hot-climate sun shield is excluded	Massing model, then weighing	Met on paper: 2.45 kg, a thin 0.05 kg margin on catalogue masses after the design for construction (FND-DDR-003); 2.61 kg with the shield
R15	Serviceable	Cell replaced in 10 min or less with a screwdriver; no soldering in the field	Design review	Met by design: about 7 min; about 9 min where the shield is fitted, since it lifts off after four thumb screws
R16	Cost	FieldNode core (enclosure, power, radio, panel and mounting) at the \$150 value-engineering target or under at quantity 1 (a hypothetical control target); sensors and gateway excluded	Priced BOM	Met on paper, within the value-engineering target: \$139.00 base node (\$11.00 under); \$148.00 with the shield (\$2.00 under)
R17	Open and independent	All design files under CERN-OHL-S-2.0 and MIT; works with any LoRaWAN network server, no closed cloud	Design review	Met by design
R18	Privacy pass-through	The core sends only what the sensor firmware passes to it; no images or audio leave the node	Firmware design review	Met by design; each project states its own rule

■ Requirements at risk

- **R2 (cold charging) at risk.** In the cold the cell charges only on clear days warmer than about -13 °C, so an overcast freeze longer than the cold autonomy will stop the node. The warm end is now covered by the shield at hot sites.
- **R6 (autonomy) at risk for a cold or aged cell.** At the published 100 mW a new cell lasts 5.75 days, but 4.03 days at -20 °C and 4.60 days at end of life.

Resolved by FND-DDR-002 on paper: R3 (shield at sites above 30 °C), R5 (shield and 9 V class panel), R9 (firmware interval rule) and R14 (base node). R3 and R5 rest on an assumed shield factor that only a test can confirm; testing is TRL 4 work and on hold.

■ Assumptions

- Assumptions for every status above are stated in FND-CAL-001, Table 1. The main ones: worst month 2 peak sun hours on the tilted panel; panel, dust and angle losses 20 %; MPPT charger 85 %; cell charge 95 %; rail converters 90 %; LiFePO4 cell 3.2 V, 6 Ah (about 19 Wh), 80 % usable.
- Uplink payload 20 bytes plus 13 bytes of LoRaWAN overhead, 125 kHz bandwidth, coding rate 4/5, 8-symbol preamble.
- Core consumption from SX1262 datasheet figures ([Semtech SX1261/2 datasheet](#)): 4.0 mWh a day.